



An extraction and novelty evaluation framework for technology knowledge elements of patents

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Abstract

Technology knowledge elements play an important role in technology innovation. However, there is still challenges about their extraction and evaluation. Traditional methods exhibit limitations in precisely linking key technologies with functions, and they usually focus on measuring the overall novelty of patent documents rather than individual technology details, leading to poor interpretability and practicality of research outcomes. In this work, we present a framework that extracts technology knowledge triples and evaluates the novelty of triples based on deep learning model. This framework first identifies key sentences that reflect innovation from patent claims and then extracts technology knowledge elements from these sentences. A novelty index is then designed to evaluate the novelty of these technology knowledge elements based on the probability of their occurrence and the similarity to existing knowledge. The experimental results demonstrate the effectiveness of the proposed method. The extracted technology knowledge elements can use to construct an innovation knowledge graph, which provides practical applications in engineering knowledge retrieval, design and innovation support.

Keywords Technology knowledge elements extraction · Novelty evaluation · Technology innovation · Patent mining

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Introduction

In the field of measuring innovation through the scientific literature, Knowledge Elements (KEs) represent the smallest units of knowledge (Wang et al., 2022). KEs serve as the foundation for knowledge management activities and are often referred to entities. A collection of KEs can represent a given technology field (Strumsky et al., 2015). The extraction and evaluation of KEs can provide a better understanding of technology development, reveal new technological trends, and infer possible areas of technological development (Yoon et al., 2017; Zanella et al., 2021). It can also help researchers to derive and evaluate new technology ideas (Hong et al., 2022), facilitate product development (Lee et al., 2020), and identify emerging technologies (Kim et al., 2019). By discovering potential technology development methods, the extraction and evaluation of KEs can help identify technology opportunities. Therefore, it can be considered a subtask of Technology Opportunity Analysis (TOA) proposed by Porter and Detampel (1995). Regarding the definition of “Method Knowledge Elements (MKEs)” as proposed by Wang et al. (2022), this paper delineates “Technology Knowledge Elements (TKEs)” as knowledge elements associated with designs, techniques, functional effects, and other technical details indicative of technological innovation. The primary focus of this paper is to extract and evaluate TKEs from patents, which represent the internal knowledge embedded within scientific documents.

In recent years, scholars have increasingly turned their attention towards the granular content within scientific documents (Zhang et al., 2023), and patent documents have been widely used to extract technical knowledge (Lee, 2021). Based on the results of patent analysis, researchers can obtain technical inspiration of designing new functions (Hong et al., 2022; Lee et al., 2020) or recommend key technical fields (Jin et al., 2021). However, existing patent analysis methods such as bibliometrics (Zhao et al., 2020; Stek et al., 2021), semantic topic clustering (Ma et al., 2021; Hu et al., 2022), and keyword extraction from patent abstracts (Lee et al., 2020; Geum & Kim, 2020) have limitations in capturing the detailed information of specific innovations, and the analysis achieved tend to be coarse-grained. In addition, these methods may not capture the relations between technical elements (Zhao et al., 2020; Altuntas et al., 2020). To summarize, existing methods for extracting or evaluating technology knowledge elements (TKEs) may enable the discovery of technological hotspots, such as promising areas or functional effects in certain technology fields. However, the conclusions drawn from bibliometric or technology topic analysis are often based on subjective summarization. And researches relevant to novelty evaluation based on patent data typically focus on measuring the novelty of patent documents rather than the novelty of specific technology details. Traditional methods exhibit limitations in precisely linking key technologies with functions, and they usually focus on measuring the overall novelty of patent documents rather than individual technology details, leading to poor interpretability and practicality of research outcomes.

To address the challenges mentioned, this paper proposes a framework based on deep learning to extract and evaluate technology knowledge elements. This framework first utilizes detailed information from unstructured data in patents, such as patent claims, to identify sentences that reflect innovation and then extract technology knowledge elements from the sentences. These elements are represented as nodes and edges in an innovation knowledge graph. A novelty index is then designed to evaluate the novelty of these triples based on the probability of their occurrence and the similarity to existing knowledge. Finally, a valid experiment is conducted by comparing the novelty annotated by three researchers with the novelty obtained by the proposed method, demonstrating the effectiveness of the

proposed approach. In addition, a comparative experiment is performed to further validate the proposed novelty evaluation method.

The contributions of this paper are as follows. First, a comprehensive framework for extracting knowledge related to technological innovation from patent claims is proposed. This framework uses the BERT model to identify sentences that reflect innovation based on the verbs associated with functional effects, and then employs the UIE model to perform key triple extraction. This process provides a data foundation for achieving fine-grained TOA analysis. Second, a triple novelty evaluation index based on graph neural networks is designed. This index enables a more objective evaluation of the novelty of combinations of technology knowledge elements. Third, an annotated dataset is constructed that can be used to validate the effectiveness of the novelty evaluation method. In summary, the proposed approach aims to facilitate the identification of novel functions and corresponding key technologies, which can benefit technology opportunity analysis by providing a more objective and comprehensive evaluation of technological innovation.

The structure of this paper is as follows: the “Literature Review” section examines the theoretical foundations and prior researches on patent mining and novelty evaluation. The “Methodology” section details the framework and methods employed in this study. The “Illustration” section presents the results of the demonstration. The “Discussion and Conclusions” section delves into the implications, strengths, and limitations of the research presented in this paper.

Literature review

Patent mining

Patent mining refers to the process of extracting valuable information and insights from patent documents. It involves the systematic analysis and interpretation of patent data to discover trends, patterns, and knowledge related to technological advancements and competitive intelligence (Abbas et al., 2014). The goal of patent mining is to gain strategic insights, identify emerging technologies, assess patent portfolios, and make informed decisions regarding R & D or innovation strategies.

A current trend is to automatically analyze patent documents using software and algorithms to extract relevant content for technological innovation (Guo et al., 2022). Some research analyze patents’ structured bibliographic items such as International Patent Classification (IPC) codes or citation information. Cho et al. (2021) employed a Decision Tree (DT) based approach to forecast potential IPC co-occurrences, aiming to predict potential instances of technology convergence. Nakai et al. (2018) employed LSTM to forecast the trend of citation network’s expand, thereby extracting the progression of technology development. Although these methods are objective, intuitive, and efficient, they merely rely on statistic information and unable to incorporate textual semantics into their analysis. Other research analyze patents’ unstructured text such as abstracts (Ren & Zhao, 2021). Yoon et al. (2019) used Generative Topology Mapping (GTM) and the Local Outlier Factor (LOF) to identify vacant technology areas in the field of Information and Communications Technology (ICT) through the key word vector of patents. Lee et al. (2020) employed a three-layer neural network to derive Word2vec word vectors from patent data and constructed a product map by calculating the similarity matrix of these word vectors, offering insights into the current state of technological advancements. Zhang et al. (2021) combined

topic vectors obtained through Linear Discriminant Analysis (LDA) model with the patents' release timeline, to discover the evolution trajectory of various technologies in the blockchain field. The extracted technical hot-spots are regarded as the most promising technical opportunities. This type of research aims to identify technological opportunities by extracting semantic information from unstructured text, allowing the acquisition of more comprehensive, profound, and intricate insights compared to traditional statistical data. It facilitates fine-grained analysis of technological aspects.

Thus, this paper intends to extract semantic information (such as TKEs) from patent texts and facilitate in-depth patent mining. In addition, most of the current related research on patent analysis is based on analyzing the full text of patent abstracts, and the amount of data processing is relatively large. Therefore, this paper conducts a sentence-level patent analysis, which may simplify the data processing task. The patent claims are long texts containing many words and complex semantic relations. Therefore, a framework is further designed to extract semantic TKEs in the patent claim, with the purpose of providing inspiration for developers and improving the value of the analysis.

Novelty evaluation

The definition of novelty varies across different contexts. In the field of news reports' novelty evaluation research, novelty refers to a report that discusses a topic for the first time in a chronological sequence of news reports (Allan et al., 1998). In the process of technology scouting and substantive patent examination, novelty generally refers to the fact that the content of the patent document has not been disclosed in whole or in part by any relevant prior art before the priority date or filing date (Verhoeven et al., 2016). In the evaluation of technological achievements, novelty refers to the fact that the research topic has not appeared in a specific retrieval dataset or exhibits significant differences compared to previous results. This concept is known as relative novelty and constitutes a primary focus of measurement in existing research.

Novelty carries different meanings depending on the specific scenarios and can be measured using various features such as contribution sentences, topic words or key words. Hasan et al. (2009) proposed a novelty evaluation method based on patent content, the time when the terms of patent were initially proposed was employed to represent patent's novelty. Arts et al. (2018) measured the similarity between technologies by calculating the Jaccard value between patent keywords, providing a method of evaluating patent novelty and finding technical opportunities. Wang et al. (2019) employed Implicit Semantic Analysis (ISA) model to extract the relation between words in patent documents, subsequently employed outlier detection methods to identify abnormal patents, considering outlier patents as novel due to their divergence from the majority of patents. De et al. (2021) calculate the novelty of the articles by matching the text with a knowledge graph. Novelty is inversely proportional to the number of common nodes between the medical concept subgraph and the corresponding subgraph of knowledge graph. A higher novelty score indicates a greater difference between the text and existing knowledge, indicating higher levels of innovation and uniqueness. Wang et al. (2022) utilized the cloud similarity equation to calculate the similarity between different types of research contribution sentences and combined contribution sentences of the same type into a cloud. A low cloud similarity, denoted by a small overlapping area, indicates a high degree of novelty. Wang et al. (2023) proposed a framework to measure the novelty of scientific literature based on research contributions. The framework

utilizes deep learning and cloud models to identify and classify contribution sentences, facilitating the rapid identification of highly creative articles within a large corpus of papers.

In summary, the meaning of novelty varies depending on the specific context and can encompass aspects such as being the first report, absolute novelty from scratch, relative novelty, and differences from existing results. When evaluating novelty, factors such as publication date and novelty of patent content can be taken into account. Based on the aforementioned method of measuring differences between knowledge elements, this paper designs a novelty index to calculate the differences between semantic combinations of TKEs.

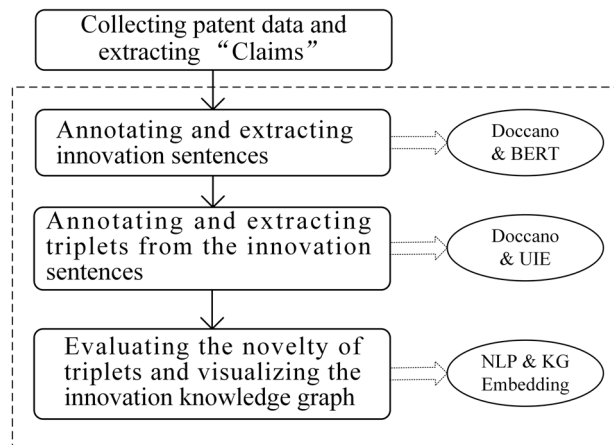
Methodology

Research framework

This paper proposes a framework for extracting and evaluating TKEs, intending to reveal innovations in specific domains and analyze potential technological opportunities. Therefore, corresponding steps are designed to achieve the above goals, as shown in Fig. 1.

After collecting patent documents, the required bibliographic item “Claims” is extracted. The doccano platform (Nakayama et al., 2018) is chosen to annotate sentences and the Bidirectional Encoder Representations from Transformers (BERT) model is fine-tuned to classify the original sentences into two groups: sentences that indicate innovation and sentences that can not indicate innovation. Then, entities and relations in sentences that represent innovation are annotated, and the Unified Structure Generation for Unified Information Extraction (UIE) model is used to extract innovation triples. Finally, the evaluation of each triple’s novelty based on Natural Language Processing (NLP) methods and Knowledge Graph Embedding methods is employed; a visualization of the result is also presented, revealing the technology hotspots. Since this article is not aimed at analyzing the path of technological evolution, the result of visualization is only presented to indicate what this knowledge element extraction and evaluation method can find out.

Fig. 1 Research framework



Innovation sentences identification based on BERT

BERT is a pre-trained model that can be used for text classification tasks (Devlin et al., 2018). In the pre-training stage, the BERT model uses the [MASK] label to cover some words. A two-way self-attention mechanism is designed to learn the semantic information of each word through predicting the missing token. And a linear classifier (in this case, a softmax layer) is trained for sentence classification.

This paper utilizes a fine-tuned BERT model to identify “innovation sentences” that indicate innovative accomplishments, making the proposed framework allow for sentence-level analysis. And the Doccano platform is used to annotate these innovation sentences. The basis for the annotation is the work of Korobkin et al. (2019), which discovered high-frequency verbs that refer to functional effects. Since innovation in technology involves substantial improvements or evolution at the technical level based on existing technologies, leading to the generation of new products, processes, and methods (Edwards-Schachter, 2018), the novelty of functional effect could be considered as a representation of the degree of innovation. Therefore, sentences identified based on verbs that indicate functional effect should be able to reflect technological innovation in a way.

The process of identifying innovation sentences could be viewed as a binary sentence classification task, achieved by using a fine-tuned BERT model and linear classifier. After extracting patent claims and segmenting the sentences, they will be sorted into two groups: “innovation sentences” and “non-innovation sentences”. Only innovation sentences will be included in subsequent analysis and used as a data source to extract key triples.

Key triples extraction based on UIE

Innovation sentences identified by the BERT-based method serve as a data source for key triple extraction. The types of entities and relations are designed based on the high-frequency verbs discovered through Subject-Action-Object analysis, and the Doccano platform is used to annotate the entities and relations. The annotated sentences are then split into three datasets (train set, dev set, and test set) to fine-tune the UIE model. Finally, the fine-tuned UIE model is used to extract triples from unannotated sentences. These extracted nodes and relations reflect the functional effects of patents and can be considered key triples in the analysis of innovation. Thus, the extraction of TKEs can be viewed as the joint extraction of key triples from innovation sentences.

UIE model is applied to implement the joint extraction of innovation entities and relations (Lu et al., 2022). UIE uses SEL (Structured Extraction Language) to describe the hierarchical relation in the training text and uses a Structural Schema Instructor (SSI) to control which information is extracted. SSI is a Schema-based Prompt engineering that guides the identification and association of specific information. By utilizing SSI, the generation of SEL can be effectively guided and semantic information can be shared among different labels and tasks.

The original UIE can uniformly generate different information extraction structures. When performing the relation extraction task, the [spot] label represents the entity, and the [asso] label represents the relation. The model first identifies different segments according to the labels in front of the sentence, namely “Spotting”, and then determines the relation between these segments, namely “Associating”, as is shown in Fig. 2. By using these labels, UIE is promoted to predict new SEL structures from sentences without any label.

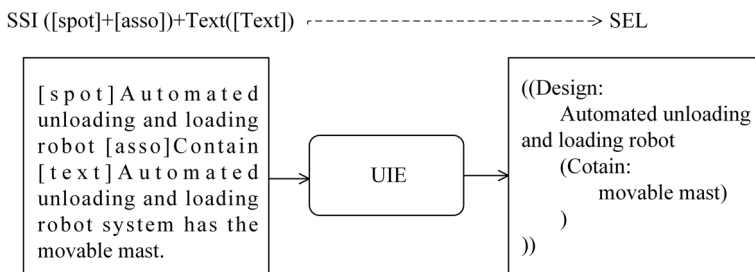


Fig. 2 Application examples of UIE in IE tasks

Specifically, assuming there is an input text sequence x and SSI s , UIE first calculates the hidden representation $H = \{s_1, \dots, s_i, x_1, \dots, x_i\}$ for each token. Then use the encoder to generate the linear SEL sequence of the target record $y = \{y_1, \dots, y_i\}$, and the decoder will decode the hidden representation and the previously generated SEL sequence to predict the token in the following SEL sequence.

The UIE algorithm used in this paper comes from the open-source code of Baidu Paddle Studio (PaddleNLP Contributors, 2021). On the basis of the original UIE paper's idea, Paddle-UIE applies the attention mechanism mentioned above and uses the Transformer-based ERNIE 3.0 model (Sun et al., 2021) to perform information extraction tasks. The difference between the paddle's UIE model and the original paper is that as a discriminative model based on a fixed prompt, Paddle-UIE performs better in zero-shot tasks and few-shot tasks with an advantage in speed. Instead of estimating the distribution of the joint probability like generative models, the discriminative model uses conditional distribution for classification (Ng & Jordan, 2001). This paper choose paddle's UIE to extract triples for the advantage metioned above.

Novelty evaluation based on HGT and TransH

The traditional novelty evaluation methods mainly focus on the whole patent level, less considering the innovation of fine-grained technical knowledge. In addition, due to the possible semantic overlap among triples of fine-grained technology knowledge triples extracted from the patent claims, the semantic complexity makes it difficult for the traditional node influence-based methods to effectively measure the novelty of the triples. To solve this problem, this paper performs novelty evaluation on the extracted triples from the perspective of semantic similarity, following the approach proposed by Luo et al. (2022).

Our proposed method defines novelty as the distinctions and uniqueness of a knowledge element in comparison to existing knowledge. When a combination of knowledge elements has a low probability of occurrence in a network, it implies a weak connection with other known combinations of knowledge elements, indicating a greater distance in the vector space (Hain et al., 2022). Thus, low occurrence probability can be interpreted as higher novelty of the triple. Furthermore, the semantic similarity of the text reveals the distinctions in semantic roles and functions within the contextual semantic environment of words, which plays an important role in reflecting novelty (Luo et al., 2022). Scholars (Arts et al., 2018; Wang et al., 2019) have used semantic similarity to assess novelty and obtained similar results that align with real-world observations. In this paper, a novelty index is developed incorporating both the combination probability of knowledge elements and semantic

similarity, reflecting the differences between specific triples and existing triples in the domain and thus capturing the novelty of the triples.

To identify triples that can reflect technological innovation from the extracted key triples, a novelty index is designed for the evaluation of novelty. Evaluating novelty implies comparing with existing knowledge. This paper considers a triple to be novel if it has a low probability of occurrence. The novelty of triples is evaluated by measuring the degree of novelty of the methods or technologies represented by them. Therefore, the lower the probability of a triple appearing in the knowledge graph, the higher its novelty is considered to be. This paper combines knowledge graph embedding models and entropy weighting methods to design a novelty evaluation method that incorporates both link probability and semantic similarity of technology knowledge elements. Figure 3 illustrates this approach.

The Heterogeneous Graph Transformer (HGT) (Hu et al., 2020) is a graph neural network designed to model large-scale heterogeneous graphs within the web domain. By preserving type-specific parameters, it enables specialized representations of different node and edge types, facilitating the modeling of the heterogeneous structure. Notably, HGT incorporates a novel attention mechanism that automatically learns important “meta-paths” without manual design, making it highly effective in handling heterogeneous graph data. This attention mechanism has shown promising results in various real-world applications. In this paper, HGT is employed to obtain representation vectors of nodes and edges in the knowledge graph, which are subsequently used in further computations.

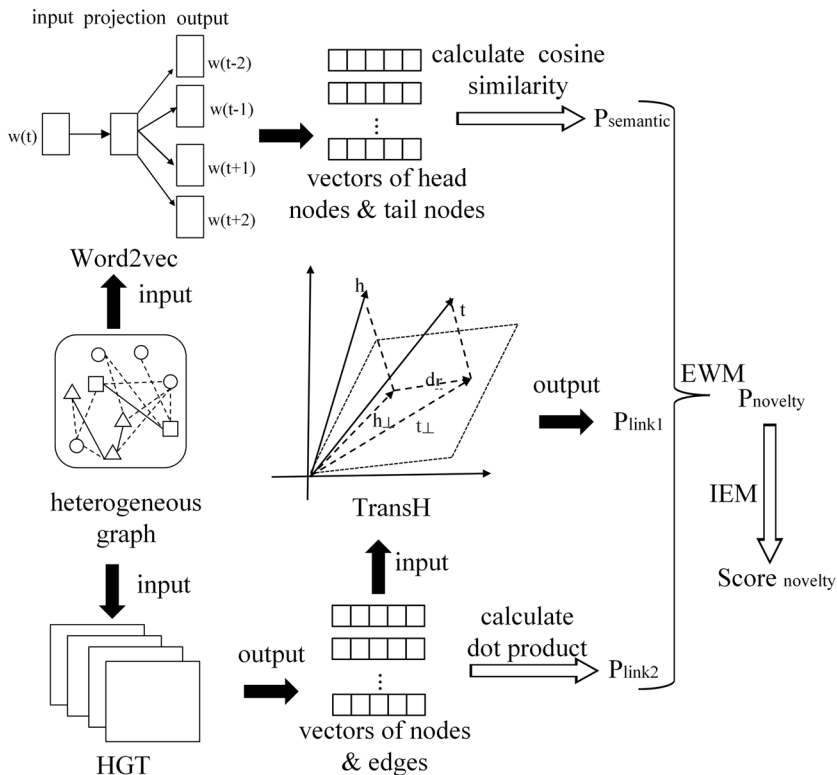


Fig. 3 The procedure of novelty evaluation

During aggregating, a first-order aggregation strategy is utilized in the convolution process to aggregate information from neighboring nodes. This choice is motivated by the need to capture local dependencies in the graph while maintaining computational efficiency. The advantage of first-order aggregation is that it avoids higher-order computations and simplifies the learning process, making it computationally feasible for large-scale heterogeneous graphs.

If a method contains combinations of knowledge elements that are unlikely to be linked, the technique is considered to be novel. Link prediction methods are used to calculate the probability of indirect links between technology knowledge elements using the vectors obtained through HGT. The vectors obtained by HGT are used as inputs to the TransH model (Wang et al., 2014) and calculates the probability of a triple being valid according to formula (5). This probability is used to represent the co-occurrence probability of the head and tail nodes.

TransH is a model that projects the entities and relations into a common space and calculates the distance or similarity between them using a scoring function. The scoring function used in TransH is typically based on the L1 or L2 distance measure. Assuming the projected head entity as $\mathbf{h}_p$, the projected tail entity as $\mathbf{t}_p$, and the projected relation as $\mathbf{r}_p$. The score function used in TransH is defined as:

$$f(\mathbf{h}_p, \mathbf{t}_p, \mathbf{r}_p) = \|\mathbf{h}_p + \mathbf{r}_p - \mathbf{t}_p\|_p \quad (1)$$

where $\|\cdot\|_p$ represents the L_p norm, and the choice of p determines the distance measure used. For example, when $p = 1$, it corresponds to the L1 distance, and when $p = 2$, it corresponds to the L2 distance.

To calculate the possibility of one edge to exist, this paper uses the vector obtained by HGT as the input of the transH model to calculate the value of P_{link1} : assuming there is a triple $(\mathbf{h}_p, \mathbf{t}_p, \mathbf{r}_p)$, the P_{link1} scores can be computed as:

$$P_{link1} = \|\mathbf{h}_{p_{HGT}} + \mathbf{r}_{p_{HGT}} - \mathbf{t}_{p_{HGT}}\|_p \quad (2)$$

This index represents the possibility for two entities to connect, a higher value of the indicator P_{link1} means that the head entity and the tail entity are more likely to form an effective triple under a given relation, and the novelty is lower. However, the probability of linking between nodes is usually a complex issue, and it may not be possible to fully capture all associated information relying on a single indicator. By considering multiple indicators in a comprehensive way, a more accurate and comprehensive estimate of the probability of the link can be obtained. Therefore, another indicator needs to be considered to capture multiple factors that affect link probability. Thus, this paper supplements the design of a second indicator P_{link2} for calculating the probability of triple formation:

$$P_{link2} = \sigma(\mathbf{h} \cdot \mathbf{r} \cdot \mathbf{t}) \quad (3)$$

where $\mathbf{h}$ represents the node vector of the head entity, $\mathbf{r}$ represents the edge vector of the relation, and $\mathbf{t}$ represents the node vector of the tail entity. The establishment probability of the triple is obtained by calculating the dot product of the head entity vector and the tail entity vector, $\sigma(\cdot)$ represents the sigmoid function, and maps the result to the range [0, 1]. This metric utilizes the dot product between vectors to measure the similarity between head entities, relations, and tail entities. The higher the value of the indicator P_{link2} is, the lower the novelty.

Lastly, if the semantic similarity between a pair of technologies knowledge elements is low, the relation between them is considered to be novel. The skipgram model of word2vec is employed to calculate the semantic similarity between each pair of knowledge elements based on the cosine similarity between the word vectors of the head and tail nodes:

$$P_{semantic} = \frac{\mathbf{h} \cdot \mathbf{t}}{\|\mathbf{h}\| \|\mathbf{t}\|} = \frac{\sum_{i=1}^n h_i \times t_i}{\sqrt{\sum_{i=1}^n h_i^2} \times \sqrt{\sum_{i=1}^n t_i^2}} \quad (4)$$

where, $\mathbf{h}$ and $\mathbf{t}$ represent the vector of head node and tail node respectively, h_i and t_i are the i -th elements of the two vectors respectively, and n is the amount of vector's elements. $P_{semantic}$ represents the cosine similarity between two vectors, ranges from -1 to 1 with a value closer to 1 indicating a higher degree of similarity, and means the novelty is lower.

The three indicators contribute differently when constructing the comprehensive novelty metric, denoted $P_{novelty}$. In this study, the entropy weighting method is employed to determine the weights for the three indicators. A higher degree of indicator dispersion corresponds to a lower entropy value, indicating a greater influence of the indicator on the overall evaluation and thus a higher weight for that particular indicator. The steps for calculating the weights based on the Entropy Weighting Method (EWM) are as follows.

First, normalize the values of each indicator to a $[0, 1]$ range:

$$p_{ij} = \frac{p_{ij} - \min(p_i)}{\max(p_i) - \min(p_i)} \quad (5)$$

where p_{ij} represents the value of the i index of the j th triple. Then calculate the entropy value for each indicator using the formula:

$$E_i = - \sum_{j=1}^n p_{ij} \log p_{ij} \quad (6)$$

Then calculate the weight (W_i) for each indicator:

$$W_i = \frac{1 - E_i}{\sum_{i=1}^m (1 - E_i)} \quad (7)$$

where E_i represents the entropy value for the i -th indicator, p_{ij} represents the normalized value of the i -th indicator for the j -th sample, n is the total number of samples, W_i represents the weight for the i -th indicator, and m is the total number of indicators.

Finally, to facilitate computation, the value of $Score_{novelty}$ is scaled up through Index Expansion Method (IEM). If the magnified novelty score is less than 0, it indicates a high level of novelty (Klavans et al., 2006). The formula for $score_{novelty}$ is:

$$Score_{novelty_i} = \frac{p_{novelty_i} - \mu}{\sigma_{var}} \quad (8)$$

where $P_{novelty_i} = \sum_{n=1}^3 (W_n P_n)$, represents the probability score of the i -th triple, $P_n = P_{link1} \cdot P_{link2} \cdot P_{semantic}$. If the value of $Score_{novelty_i}$ is less than 0, then this triple has high novelty.

Experiments and results

Data collection and preprocessing

Agricultural robots are automated devices used in agricultural production. They can significantly improve agricultural production efficiency, reduce labor costs, save energy and reduce environmental pollution in agricultural production (Zhao et al., 2020). Agricultural robots may become a vital force in planting and farming. However, current research on agricultural robots is still in the laboratory stage, and it is still difficult to apply fully automatic and precise control on robots in practical applications (Jin et al., 2021).

Based on the fact that agricultural robots are capable of automating many agrarian tasks such as planting, watering, fertilizing, harvesting, and packaging agricultural products. This paper designs a query set using keywords like “mechanical arm”, “agricultural robot”, “harvest robot” and so on, as shown in Table 1. Then the query set is used to collect patents of USPTO from the “Zhihuiya” website. In total, 1343 patents of the search result are related to agricultural robots and are selected for subsequent analysis.

The patent claim is a detailed document that describes the technical details of an invention which could be considered as technology knowledge elements. As the original text data is unstructured, preprocessing steps, such as segmentation of relevant paragraphs, are carried out to prepare the data for a deep learning model. A total of 40,113 sentences are obtained for further analysis.

Innovation sentences identification

Patent claims contain valuable information about new technologies, making them an important source for technology analysis and innovation research. Based on the work of Korobkin et al. (2019), who extracted descriptions of physical effects from patent claims in the USPTO database, predicates such as “change”, “increase”, “decrease”, “depend”, “generate”, “act”, and “cause” are considered suitable for describing the technical benefits of patents. These verbs reflect significant functional improvements from the old technology to the current patent and can be considered as evidence of technology innovation. Therefore, sentences in patent claims that contain these words are considered “innovation sentences”, meaning that they contain technology knowledge elements and can reveal innovation. Then the BERT model is utilized to extract such innovation sentences.

Table 1 The query set and amount of relevant patents

Query set	Result
((((TACD:(agriculture*) AND (TACD:(crops) OR TACD:(fruits) OR TACD:(vegetable*) OR TACD:(seeding*) OR TACD:(harvest*))) AND (TACD:(robot*) OR TACD:(manipulator*) OR TACD:(mechanical arm*) OR TACD:(mechanical hand*) OR TACD:(mechanical hands*) OR TACD:(machanical arms*))) AND (IPC:(A01B*) OR IPC:(A01C*) OR IPC:(A01D*) OR IPC:(A01F*) OR IPC:(A01G*) OR IPC:(A01M21*) (B25J*) OR IPC:)) OR (TACD:(Agricultural robot*)) NOT (IPC:(A01G05*) OR IPC:(A01G23*))	1343

Table 2 Distribution of two types of sentences in sentences classification datasets

Index	Dataset	Innovation sentences	Non-innovation sentences
1	Train	1804	15,772
2	Dev	600	5257
3	Test	600	5258

Table 3 The performance of BERT and other models in the innovation sentences classification task

Index	Model	acc
1	BERT	97.7
27	FastText	89.8
3	Logistic Regression	90.3
4	TextCNN	89.7
5	TextRNN	91.1

A total of 40,113 sentences are collected, out of which 29,291 sentences are annotated. Among these annotated sentences, there are 3004 innovation sentences. To construct the dataset, a 6:2:2 ratio is used, as shown in Table 2, and four models, namely FastText, Logistic Regression, TextCNN, and TextRNN, are compared through experimental analysis. As shown in Table 3, the results on the self-constructed dataset demonstrated that BERT produced the best performance among the models. In the end, a total of 4150 innovation sentences are obtained.

In total, 29,291 sentences are annotated based on the verbs mentioned above. These labeled sentences are divided into training set, validation set, and test set according to the ratio of 6:2:2. Then Google's Bert-base-uncased model is applied in the environment of python 3.8.16, Tensorflow 2.6.2 and CUDA 11.2 to perform innovation sentence recognition, a binary classification task. After multiple rounds of iterative adjustment, the model achieved a F1 value of 0.97 on the test set.

Then the fine-tuned model is utilized to classify the remaining 10,822 sentences into the appropriate category. The unannotated samples are input into the model to complete the task of identifying innovative sentences. Based on the identification results, every patent contains innovation sentences, indicating that the self-constructed dataset has achieved comprehensive coverage of relevant patents. In total, 4209 innovation sentences were obtained, with an average of 3.13 innovation sentences per document.

Key triples extraction

Extracting valuable information from patent claims is crucial for TOA analysis and innovation research. However, the innovation sentences identified from patent claims are often quite long, which may make it difficult to conduct text analytics from them. Therefore, these innovation sentences are further utilized to extract triples consisting of TKEs such as entities and their relations. However, designing entity and relation types objectively and precisely remains a challenge. In this paper, a syntactic dependency analysis is conducted on innovation sentences to identify frequently occurring verbs. The top 50 verbs with the

Table 4 Top 50 verbs with the highest frequency

Word	F	Word	F	Word	F	Word	F	Word	F
Configure	3076	Generate	2967	Base	2313	Determine	1397	Cause	1204
Receive	1141	Accord	957	Have	921	Perform	860	Use	829
Correspond	741	Transmit	726	Associate	722	Control	656	Select	653
Couple	640	Detect	626	Provide	617	Store	563	Identify	549
Move	534	Execute	485	Reduce	461	Change	440	Obtain	412
Set	410	Increase	410	Apply	378	Connect	366	Capture	334
Acquire	263	Extend	261	Define	260	Adjust	254	Calculate	253
Operate	248	Desire	246	Contain	242	Sense	223	Compare	222
Collect	216	Dispose	214	Compute	213	Relate	211	Grow	211
Position	197	Update	195	Travel	195	Mount	195	Process	192

Table 5 Definition of entity types

Entity type	Definition	Label	Concrete example
Design	Patent title/system name	DE	Power assist robot
Technology	Technology	TE	Machine learning
Condition	Working \environmental condition	CO	Luminance levels
Problem	Problem that need to be solved	PR	Interference impact
OutputInformation	System's output	OI	Schedule for irrigation
Function	Function	PR	Vibration pattern analysis
BenefitIndicator	Higher value is better	BI	Signal-to-noise ratio
CostIndicator	Lower value is better	CI	Vision-disturbing characteristic
Basis	Prerequisites of function	BA	Cloud-based data

highest frequency are shown in Table 4. Based on the subject-action-object (SAO) structure extracted by the StanfordNLP model, entity and relation types are designed.

Special attention is given to the predicates mentioned during the annotation of innovation sentences, such as “*change*”, “*increase*”, “*decrease*”, “*depend*”, “*generate*”, “*act*” and “*cause*” as well as their synonyms. For example, if “*increase*” appeared frequently with subjects representing technology-related nouns and objects representing nouns indicating benefits like “*yield*”, the head node could be designated as an entity type “*Technology*”, and the relation type of the tail node could be set as “*BenefitIndicator*”. For other predicates, the appropriate entities and relation types are determined based on their specific meanings and the context. The meanings of different entities and relation types are provided in Tables 5 and 6.

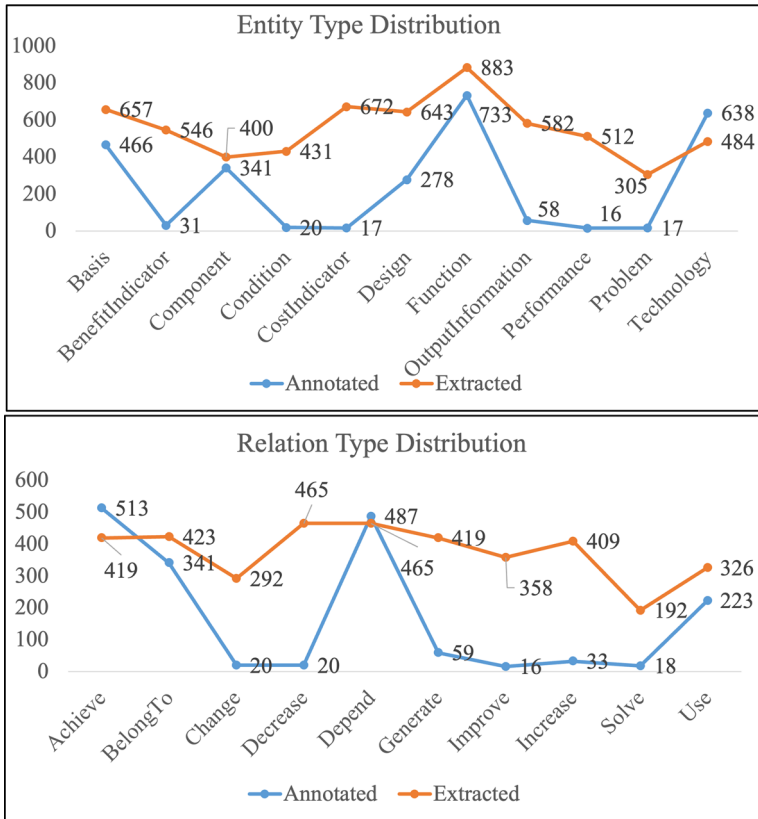
Since this paper does not aim to accomplish a comprehensive analysis of agricultural robotics patents, a subset of 2314 short sentences (with lengths below 512) from 779 patents is selected for further extraction in order to simplify the annotation process. Among these, 198 sentences are manually annotated, and 402 long sentences (with lengths over 512) are also annotated, resulting in 1730 relations and 2615 entities. The HGT model is utilized to obtain the vector of the annotated triples, and the probability of the triples' occurrence is calculated based on the TransH method, using formula (2) and formula (5). As this index reflects the possibility of the triple being accurate, it can

Table 6 Definition of relation types

Head node type	Relation type	Label	Tail node type	Definition
Design	Use	US	Technology	Indicating what technology is used in the invention
Design	Change	CH	Condition	Indicating what working or environmental condition has been changed
Component	BelongTo	BT	Design	Indicating what mechanical components/subsystems does the invention have
Technology	Achieve	AC	Function	Indicating what function does the technology achieved
Technology	Solve	SO	Problem	Indicating what problem does the technology solved
Function	Increase	IN	BenefitIndicator	Indicating what benefit can the function bring
Function	Decrease	IN	CostIndicator	Indicating what cost can the function decrease
Technology	Generate	GE	OutputInformation	Indicating what information is generated or output by the technology
Function	Depend	DP	Basis	Indicating the prerequisites of this function

Table 7 The performance of the fine-tuned UIE model

Index	Indicator	Indicator value
1	Precision	0.915
2	Recall	0.705
3	F1	0.796


Fig. 4 Distribution of relation and entity types in annotated and extracted triples

serve as a measure of the quality of the annotation, while higher values indicate higher accuracy of the annotation. The resulting precision is about 0.82, indicating a relatively high level of accuracy in the annotation.

The annotated sentences have been categorized into three datasets: training set, validation set, and test set, in an 8:1:1 ratio. The UIE model is trained on the training set and fine-tuned using the validation set, achieving an F1 score of 0.796 on the test set. The fine-tuned model is then utilized to extract triples from 2115 unannotated sentences, and the results and statistics information are presented in Table 7 and Fig. 4. The evaluation of the UIE model shows that it has a precision of about 0.915, indicating its accurate identification of correct triples. Additionally, the model achieve a

recall value at around 0.705, suggesting its ability to discover the majority of correct triples. The UIE model is effective in identifying and extracting triples from unannotated innovation sentences. And as depicted in Fig. 4, the fine-tuned UIE model shows better performance while extracting the relation type of “Change”, “Decrease”, “Generate”, “Improve”, “Increase” and “Solve”, and also performs better on the related entity types.

The extraction of triples could be considered a subtask of constructing a knowledge graph from patent claims, which is beneficial for TOA analysis. After manually merging and deleting entities and relations, a total of 5674 entities and 5497 relations are obtained. The amount of each type of entity and the relation of the annotated and extracted triples is presented in Table 8. As shown in this table, the “Function” type of entity has the highest percentage among all nodes, followed by “Basis”, which is also relevant to the achievement of the function. In terms of the count of different types of relations, “Achieve” and “Depend” have high percentages, which are also relevant to the functional effect of technology and the method or procedure to fulfill the functions. This indicates that the conducted knowledge graph mainly focuses on the functional effect of innovation, which is helpful for searching for solutions to technical problems and revealing technical hotspots through the visualization of the knowledge graph. Additionally, the knowledge graph provides a data source for novelty evaluation.

Novelty evaluation

The key triples extracted through UIE consist of TKEs that represent functions, designs, technologies, etc. However, not all key triples can reflect technological innovation, as some triples are common in patent documents and cannot distinguish one patent from another in the same field. Therefore, the proposed novelty evaluation method is applied to identify novel triples that reflect both functional effects and the novelty of technological innovation. Figure 5 shows the number of each relation type among all novel triples.

To validate the effectiveness of the proposed novel evaluation method, the proposed novel evaluation method is compared with existing evaluation method. A validation dataset is constructed manually to assist in testing the accuracy of the proposed method, and the comparative experiment and an annotation consistency test are utilized to ensure the feasibility of manual evaluation. The valid data set consisting of 310 triples is annotated by

Table 8 Counts of different entity and relation types in all triples

Entity type	Amount	Relation type	Amount
Basis	680	Achieve	930
BenefitIndicator	395	BelongTo	764
Component	481	Change	312
Condition	270	Decrease	485
CostIndicator	432	Depend	951
Design	677	Generate	478
Function	1177	Improve	374
OutputInformation	436	Increase	444
Performance	334	Solve	210
Problem	186	Use	549
Technology	606		

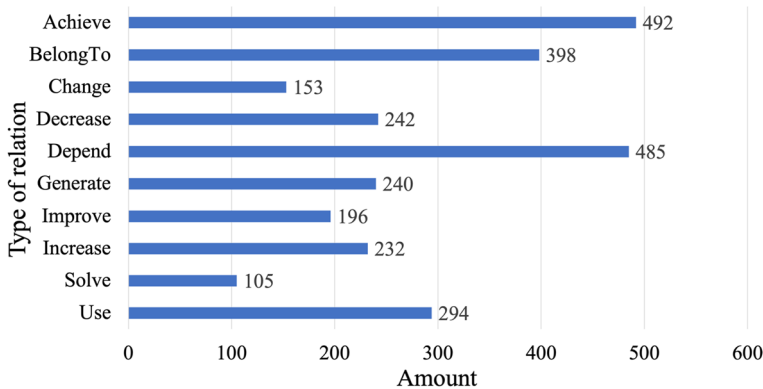


Fig. 5 Distribution of relation types in novel triples

three researchers. The ratio of each relation type is consistent between the valid triples and all triples.

The ultimate goal of agricultural robots is to enable automation in agriculture, which requires the development of new functionalities that align with current market demands (Saleem et al., 2021). Evaluating the novelty of an invention involves considering whether it incorporates previously untapped technical knowledge within the field (Verhoeven et al., 2016). These robots should perform professional tasks effectively while meeting user requirements, including maintaining a steady drive and intelligent obstacle avoidance (Tang et al., 2023). To ensure an accurate evaluation, the rough annotation criteria are developed collaboratively with the annotators, as illustrated in Table 9. Then the annotators received initial training and subsequently independently carried out the annotation process.

The annotation yields a mean Kappa consistency value of 0.77, indicating strong agreement, as shown in Table 10. The Kappa consistency value of the novelty annotation result between annotator₁ and annotator₂, as well as annotator₁ and annotator₃, is above 0.75, indicating a strong level of consistency. The Kappa consistency value of the novelty annotation result between annotator₂ and annotator₃ is greater than 0.70 but less than

Table 9 The criteria and examples for novel triples annotation

Criteria	Concrete example	Novel or not
A. Reflecting new function that was previously uncommon	a ₁ .(H)camera device-< Achieve>- (T)compare a professional activity with a user activity	YES
	a ₂ .(H)sensor-< Achieve>- (T)monitor temperature	NO
B. Reflecting novel technologies such as new technological applications, manufacturing processes	b ₁ .(H)automated action-selection system-< Use>- (T)meta prediction	YES
	b ₂ .(H)machine vision system Use>- (T) processor	NO
C. Reflecting the improvement of user experience	c ₁ .(H)analysis response circuit -< Achieve>- (T)the rebalancing	YES

Table 10 Results of annotation consistency test

Annotator	Kappa value	Asymptotic standardized error ^a	Approximate significance
Annotator ₁ -annotator ₂	0.806	0.034	0.000
Annotator ₁ -annotator ₃	0.760	0.037	0.000
Annotator ₂ -annotator ₂	0.731	0.039	0.000

0.75, indicating a moderate level of consistency. The approximate significance of the three Kappa tests is less than 0.05, with a value of 0.000, indicating that the Kappa tests are all valid.

Using formula (10), the novelty score of each triple was calculated, achieving an accuracy of 72% compared to the 310 triples manually annotated. This result demonstrates the effectiveness of the proposed method.

In addition, the method proposed in this paper is compared with the baseline method introduced by Wang et al. (2019). The novelty of triples is measured using the Angle Based Outlier Factor (ABOF) method, which calculates the angle difference between pairs of triples' text vector. A lower ABOF value indicates a larger angle difference, which may be an outlier but novel triple. The baseline method achieved an accuracy of 51%, as shown in Table 11, which is much lower than the proposed method.

To evaluate the significance of the difference between the results obtained from the two different methods, an independent samples t-test is conducted. The analysis find that the variances of the calculation results of the proposed method and ABOF method are unequal with a significance level below 0.001, indicating that the difference between the two samples is significant. This suggests that the novelty calculated by the method proposed in this paper is quite different from the novelty calculated by the ABOF method. Therefore, the novelty evaluation method used in this experiment is reliable to some extent.

Finally, the results obtained from the proposed method showed a significant correlation with the ABOF method (Wang et al., 2019). The correlation coefficient was found to be 0.360 with a p-value of 0, indicating a significant correlation at the 0.01 level. These findings provide evidence for the effectiveness of the novel novelty calculation method proposed in this paper. Then the 2837 triples identified as “novel” are stored in a Neo4j database to visualize the extraction and evaluation results of TKEs for further analysis. And to evaluate the quality of the constructed knowledge graph, this paper employed formulas (2) and (5) to generate embedding vectors for the nodes and edges and subsequently calculated the probability of triple occurrences. The obtained average probability of 0.814 indicates a relatively high level of accuracy in the constructed knowledge graph.

Moreover, the knowledge graph developed in this paper is continuously expanded by regularly collecting the latest patent data. Well-trained models will be used to extract,

Table 11 Result of the comparative experiment

Method	Counts of novel triples	Counts of non-novel triples	Accuracy (%)
Annotated by annotators	145	165	–
Proposed method	140	170	72
ABOF	66	244	51

evaluate, and store the triples, ensuring ongoing enrichment of the knowledge graph. Given the dynamic nature of patent data, the novelty of the triples is subject to change. The knowledge graph created based on these triples can also be referred to as an efficacy knowledge graph.

Application of innovation knowledge graph

The innovative knowledge graph can serve as an infrastructure for broad uses and applications in engineering knowledge retrieval, design and innovation support. For example, the innovation knowledge graph can be used to search the relevant knowledge elements around them according to semantic relationships for learning and enhancing design ideas. Such semantic relationships can also contribute to the query for solutions to specific engineering design problems or functions. In addition, clustering of nodes within the knowledge graph can yield functional technical solutions.

Figure 6 illustrates the application procedure of the innovation knowledge graph and its role in TOA analysis. With the constructed knowledge graph, users can retrieve information from the KG (Knowledge Graph) database and identify Problem-Solution triples, such as (H)Technology-<Solve>-(T)Problem or (H)Technology-<Achieve>-(T)Function, facilitating the formulation of solutions for specific functions or technical problems. Additionally, users can utilize community detection algorithms to identify significant nodes or clusters, which may represent current technology hotspots. This approach can also reveal functional technical solutions by identifying shared approaches to similar problems within each cluster. Figure 7 shows examples of the triples that reflect Problem-Solution. Figure 8 shows an example of the relatively important cluster. The meanings of each triples are shown in Table 12.

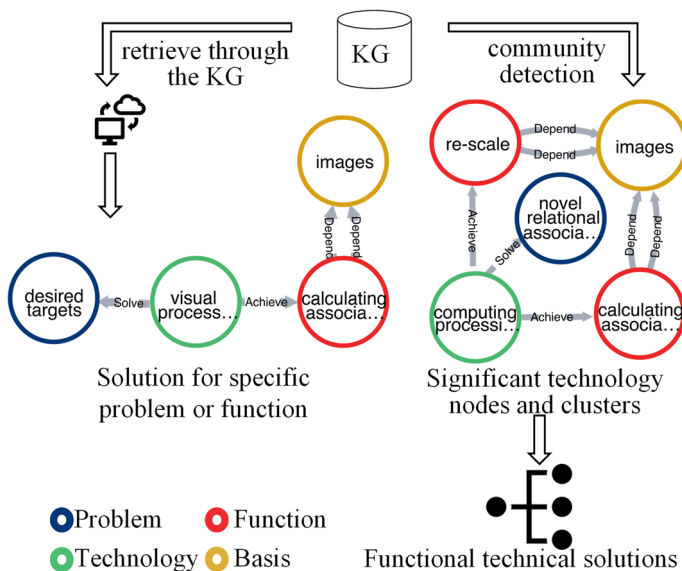


Fig. 6 Application of knowledge graph

Fig. 7 Visualization of P-S related triples in the innovation knowledge graph

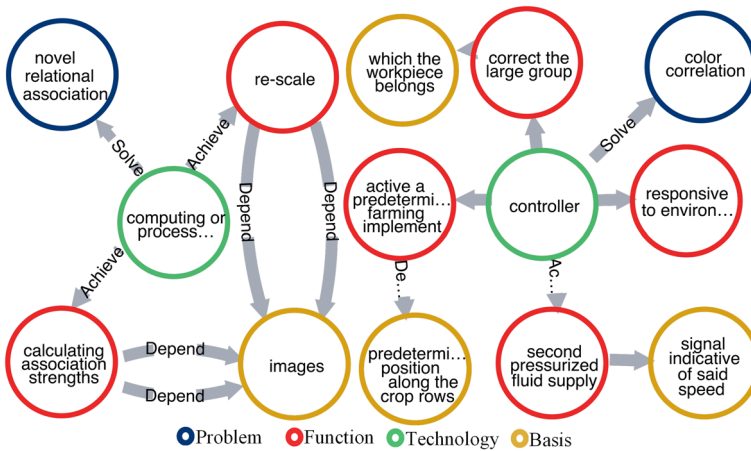
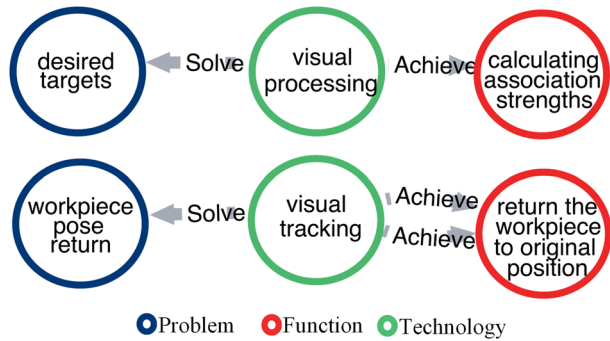


Fig. 8 Visualization of typical cluster in the innovation knowledge graph

As can be seen from the Fig. 7 and Table 12, if users retrieve issues regarding the identification of “desired targets”, the knowledge graph would provide a recommended solution, suggesting the utilization of advanced “visual processing” algorithms and techniques to process visual data and accurately detect and recognize the “desired targets”. Furthermore, it achieves the function of “calculating association strengths”, which is crucial to analyzing relationships between different elements. This calculation relies on “images” as the basis, where the “visual processing” algorithms extract relevant features and information. If users encounter issues related to the “workpiece pose return” problem, the knowledge graph would provide a retrieval result suggesting the utilization of “visual tracking” algorithms and sensors. By employing “visual tracking” technology, it can effectively track the movements and positions of the workpiece in real-time. This enables the system to accurately determine the original posture of the workpiece, ensuring precise positioning and alignment. Then, based on the workpiece’s initial posture, the system achieves the function of “returning the workpiece to its original position”, ensuring the accurate restoration of its original state and making it ready for the next operation. By retrieving from the knowledge graph, users can obtain P-S related triples that reflect potential solutions to specific problems, such as enhancing the

Table 12 Example triples in the innovation knowledge graph

(type)Head node	Relation	(type)Tail node
(Technology) visual processing	Solve	(Problem) desired targets
(Technology) visual processing	Achieve	(Function) calculating association strengths
(Function) calculating association strengths	Depend	(Basis) images
(Technology) visual tracking	Solve	(Problem) workpiece pose return
(Technology) visual tracking	Achieve	(Function) return the workpiece to an original position
(Function) return the workpiece to an original position	Depend	(Basis) original posture of the workpiece
(Technology) computing or data processing	Solve	(Problem) novel relational association
(Technology) computing or data processing	Achieve	(Function) re-scale
(Technology) computing or data processing	Achieve	(Function) calculating association strengths
(Function) re-scale	Depend	(Basis) images
(Function) calculating association strengths	Depend	(Basis) images
(Technology) controller	Achieve	(Function) active a predetermined farming implement
(Technology) controller	Achieve	(Function) responsive to the environmental data
(Technology) controller	Achieve	(Function) correct the large group
(Technology) controller	Achieve	(Function) second pressurized fluid supply
(Technology) controller	Solve	(Problem) color correlation
(Function) active a predetermined farming implement	Depend	(Basis)predetermined position along the crop rows
(Function) responsive to the environmental data	Depend	(Basis) activate a vapor generation device
(Function) correct the large group	Depend	(Basis) which the workpiece belongs
(Function) second pressurized fluid supply	Depend	(Basis) signal indicative of said speed
(Function) second pressurized fluid supply	Depend	(Basis) wheeled vehicle

lifting of robot arms and tracking the position of the robot during performing an agricultural task automatically.

As can be seen from the Fig. 8 and Table 12, in the cluster of the “*computing or data processing*” technology, the main problem addressed is establishing “*novel relational associations*”. This technology utilizes various “*computing or data processing*” techniques to analyze and extract meaningful patterns and relationships from complex datasets. It achieves the function of “*re-scale*”, enabling the transformation and normalization of data to facilitate meaningful comparisons and analysis. Moreover, it also plays a crucial role in “*calculating association strengths*”, which involves quantifying the relationships and linkages between different elements. Both the “*re-scale*” and “*calculating association strengths*” functions depend on “*images*” as the basis, where computational algorithms process the image data to derive valuable insights and measurements.

The innovation knowledge graph can also be used to identify the technology hotspot. Figure 8 shows triples related to the “*controller*”. The nodes surrounding the “*controller*” nodes form an aggregated visual effect, highlighting its importance in agricultural robotics, indicating that it is a technical hotspot in the field, and research about controllers could be a technological opportunity for the field. The detailed information of each triple is shown in Table 12. In the field of agricultural robots, the controller is widely used, particularly in achieving the function of obstacle avoidance (Liu et al., 2019). The controller’s ability to interpret sensor inputs and provide precise control commands enables the robot to navigate complex agricultural environments with precision and reliability. As shown in Table 12 and Fig. 8, the controller can perform predetermined operations, “*activate implements*” according to “*predetermined positions along the crop row*”, provide “*second pressurized fluid supply*” based on “*signal indicative of said speed of wheeled vehicle*”, “*correct the large group of pixels*” based on “*which the workpiece belongs*” and “*responsive to environmental data*”. These designs in the controller function enable agricultural robots to adapt to changing environmental conditions, such as weather and soil, and adjust actions based on the workpiece’s characteristics. It can be concluded from this cluster that improving controller performance through advancements in information processing, storage, and instruction generation is a promising research direction in the field of agricultural robots. Furthermore, as shown in Table 13, by summarizing the functional clusters, users can obtain functional technical solutions for similar problems.

In summary, by retrieving information from the innovation knowledge graph, users can obtain potential solutions for specific technical problems. Additionally, by employing clustering algorithms, such as community detection, within the innovation knowledge graph, users can observe the arrangement and interactions among Technical Knowledge Elements (TKEs). This facilitates the identification of promising directions

Table 13 The examples for functional technical solutions

Function/problem cluster	Solution
1. Digging association between images	Get novel relational association; re-scale
2. Color correlation	Correct the large group of pixels based on which the workpiece belongs; image recognition ; the likelihood as the set ratio is biased ; observation regions having a small sum total of the weights
3. Adjusting operating state of the irrigation vehicle	Electively start, stop, and regulate a flow speed based on stream of treated droplets

for technical development. Moreover, users can also obtain functional solutions for similar technical problems through the clustering process.

Discussion and conclusions

Our approach focuses on extracting and evaluating the technology knowledge elements with functionally-meaningful semantic associations from patent claims. By contrast, the previous study only mined patent titles and abstracts, and the type of knowledge is rather monotonous. Our work also contributes to a more comprehensive understanding of technological landscapes. The generated triples of technology knowledge elements could input into Neo4j to construct an innovation knowledge graph, which can serve as a valuable tool for guiding future innovations and identifying technology hotspots and trends for researchers in the field.

In present work we utilize advanced deep learning techniques and knowledge graph embedding models to identify and evaluate technological innovation in a more effective and objective manner. However, there are great potentials to derive even better-performing method in future research via improving the quality of training data, fine tuning the procedures, and exploring alternative techniques, as well as expanding the application of innovation knowledge graph.

Firstly, due to the segmented structure of patent claims, resulting in semantic segmentation in content, our study extracts sentences individually from each paragraph, inadvertently ignores a lot of semantics. A potential enhancement could involve segmenting patent claims based on semantic criteria before sentence extraction, which would contribute to obtaining sentences that are more semantically comprehensive. Secondly, the types of nodes and relations of technology knowledge triples could be better designed based on more comprehensive pre-work, such as conducting more detailed SAO analysis or using expert-summarized guide examples. In addition, our approach exclusively evaluate the novelty of technology knowledge triplets through semantic similarity. Nonetheless, given the intricacies of semantics and the heterogeneity present in patent claim formulations, it may not achieve precise and comprehensive evaluation of the novelty of technology knowledge. In the future, exploring the incorporation of some network structural factors on top of semantic similarity could enhance the robustness of this method and broaden its applicability to diverse patent contexts. The third is to continually enhance and evaluate the quality of innovation knowledge graph by incrementally updating more technology knowledge elements and involving more human expert evaluators. This will facilitate the exploration of broader applications for our constructed knowledge graph.

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Declarations

Conflict of interest The authors declare that they have no conflicts of interest that are relevant to the content of this article.

References

- Abbas, A., Zhang, L., & Khan, S. U. (2014). A literature review on the state-of-the-art in patent analysis. *World Patent Information*, 37, 3–13. <https://doi.org/10.1016/j.wpi.2013.12.006>
- Allan, J., Carbonell, J. G., Doddington, G., et al. (1998). Topic detection and tracking pilot study final report. *Carnegie Mellon University*. <https://doi.org/10.1184/R1/6626252.v1>
- Altuntas, S., Erdogan, Z., & Dereli, T. (2020). A clustering-based approach for the evaluation of candidate emerging technologies. *Scientometrics*, 124(2), 1157–1177. <https://doi.org/10.1007/s11192-020-03535-0>
- Arts, S., Cassiman, B., & Gomez, J. C. (2018). Text matching to measure patent similarity. *Strategic Management Journal*, 39(1), 62–84. <https://doi.org/10.1002/smj.2699>
- Cho, J. H., Lee, J., & Sohn, S. Y. (2021). Predicting future technological convergence patterns based on machine learning using link prediction. *Scientometrics*, 126, 5413–5429. <https://doi.org/10.1016/j.wpi.2013.12.006>
- Contributors, P. (2021). Paddlenlp: An easy-to-use and high performance nlp library. <https://github.com/PaddlePaddle/PaddleNLP>
- Devlin, J., Chang, M. W., & Lee, K. et al. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* <https://doi.org/10.48550/arXiv.1810.04805>
- de Diego, I. M., González-Fernández, C. & Fernández-Isabel, A. et al. (2021). System for evaluating the reliability and novelty of medical scientific papers. *Journal of Informetrics* 15(4):101,188. <https://doi.org/10.1016/j.joi.2021.101188>
- Edwards-Schachter, M. (2018). The nature and variety of innovation. *International Journal of Innovation Studies*, 2(2), 65–79. <https://doi.org/10.1016/j.ijis.2018.08.004>
- Geum, Y., & Kim, M. (2020). How to identify promising chances for technological innovation: Keygraph-based patent analysis. *Advanced Engineering Informatics*, 46(101), 155. <https://doi.org/10.1016/j.aei.2020.101155>
- Guo, L., Yan, F., Li, T., et al. (2022). An automatic method for constructing machining process knowledge base from knowledge graph. *Robotics and Computer-Integrated Manufacturing*, 73(102), 222. <https://doi.org/10.1016/j.rcim.2021.102222>
- Hain, D. S., Jurowetzki, R., Buchmann, T., et al. (2022). A text-embedding-based approach to measuring patent-to-patent technological similarity. *Technological Forecasting and Social Change*, 177(121), 559. <https://doi.org/10.1016/j.techfore.2022.121559>
- Hasan, M.A., Spangler, W.S. & Griffin, T., et al. (2009). Coa: Finding novel patents through text analysis. In: *Proceedings of the 15th ACM SIGKDD international conference on Knowledge discovery and data mining*, pp 1175–1184. <https://doi.org/10.1145/1557019.1557146>
- Hong, S., Kim, J., Woo, H. G., et al. (2022). Screening ideas in the early stages of technology development: A word2vec and convolutional neural network approach. *Technovation*, 112(102), 407. <https://doi.org/10.1016/j.technovation.2021.102407>
- Hu, R., Ma, W., Lin, W., et al. (2022). Technology topic identification and trend prediction of new energy vehicle using lda modeling. *Complexity*. <https://doi.org/10.1155/2022/9373911>
- Hu, Z., Dong, Y., Wang, K., et al. (2020). Heterogeneous graph transformer. *Proceedings of the web conference*, 2020, 2704–2710. <https://doi.org/10.1145/3366423.3380027>
- Jin, Y., Liu, J., Wang, X., et al. (2021). Technology recommendations for an innovative agricultural robot design based on technology knowledge graphs. *Processes*, 9(11), 1905. <https://doi.org/10.3390/pr9111905>
- Kim, K. H., Han, Y. J., Lee, S., et al. (2019). Text mining for patent analysis to forecast emerging technologies in wireless power transfer. *Sustainability*, 11(22), 6240. <https://doi.org/10.3390/su11226240>
- Klavans, R., & Boyack, K. W. (2006). Identifying a better measure of relatedness for mapping science. *Journal of the American Society for Information Science and Technology*, 57(2), 251–263. <https://doi.org/10.1002/asi.20274>
- Korobkin, D., Shabanov, D., & Fomenkov, S., et al. (2019). Construction of a matrix “physical effects–technical functions” on the base of patent corpus analysis. In: *Creativity in Intelligent Technologies and Data Science: Third Conference, CIT & DS 2019, Volgograd, Russia, September 16–19, 2019, Proceedings, Part II 3*, Springer, pp 52–68. https://doi.org/10.1007/978-3-030-29750-3_5
- Lee, C. (2021). A review of data analytics in technological forecasting. *Technological Forecasting and Social Change*, 166(120), 646. <https://doi.org/10.1016/j.techfore.2021.120646>
- Lee, C., Jeon, D., Ahn, J. M., et al. (2020). Navigating a product landscape for technology opportunity analysis: A word2vec approach using an integrated patent-product database. *Technovation*, 96(102), 140. <https://doi.org/10.1016/j.technovation.2020.102140>

- Liu, Z., Lü, Z., Zheng, W., et al. (2019). Design of obstacle avoidance controller for agricultural tractor based on ros. *International Journal of Agricultural and Biological Engineering*, 12(6), 58–65. <https://doi.org/10.25165/j.ijabe.20191206.4907>
- Lu, Y., Liu, Q., & Dai, D., et al. (2022). Unified structure generation for universal information extraction. arXiv preprint [arXiv:2203.12277](https://arxiv.org/abs/2203.12277)<https://doi.org/10.48550/arXiv.2203.12277>.
- Luo, Z., Lu, W., Cai, L., et al. (2022). Application of lexical functions in novelty measurement of academic papers. *Journal of the China Society for Scientific and Technical Information*, 41(7), 720–732. <https://doi.org/10.3772/j.issn.1000-0135.2022.07.006>
- Ma, T., Zhou, X., Liu, J., et al. (2021). Combining topic modeling and sao semantic analysis to identify technological opportunities of emerging technologies. *Technological Forecasting and Social Change*, 173(121), 159. <https://doi.org/10.1016/j.techfore.2021.121159>
- Nakai, K., Nonaka, H., Hentona, A., et al. (2018). Community detection and growth potential prediction using the stochastic block model and the long short-term memory from patent citation networks. In: *2018 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*, IEEE, pp 1884–1888. <https://doi.org/10.1109/IEEM.2018.8607487>.
- Nakayama, H., Kubo, T., & Kamura, J., et al. (2018). doccano: Text annotation tool for human, software available from <https://github.com/doccano/doccano>.
- Ng, A., & Jordan, M. (2001). On discriminative vs. generative classifiers: A comparison of logistic regression and naive bayes. *Advances in neural information processing systems* 14.
- Porter, A. L., & Detampel, M. J. (1995). Technology opportunities analysis. *Technological Forecasting and Social Change*, 49(3), 237–255. [https://doi.org/10.1016/0040-1625\(95\)00022-3](https://doi.org/10.1016/0040-1625(95)00022-3)
- Ren, H., & Zhao, Y. (2021). Technology opportunity discovery based on constructing, evaluating, and searching knowledge networks. *Technovation*, 101(102), 196. <https://doi.org/10.1016/j.technovation.2020.102196>
- Saleem, M. H., Potgieter, J., & Arif, K. M. (2021). Automation in agriculture by machine and deep learning techniques: A review of recent developments. *Precision Agriculture*, 22, 2053–2091. <https://doi.org/10.1007/s11119-021-09824-9>
- Stek, P. E. (2021). Identifying spatial technology clusters from patenting concentrations using heat map kernel density estimation. *Scientometrics*, 126(2), 911–930. <https://doi.org/10.1007/s11192-020-03751-8>
- Strumsky, D., & Lobo, J. (2015). Identifying the sources of technological novelty in the process of invention. *Research Policy*, 44(8), 1445–1461. <https://doi.org/10.1016/j.respol.2015.05.008>
- Sun, Y., Wang, S., & Feng, S., et al. (2021). Ernie 3.0: Large-scale knowledge enhanced pre-training for language understanding and generation. arXiv preprint [arXiv:2107.02137](https://arxiv.org/abs/2107.02137)<https://doi.org/10.48550/arXiv.2107.02137>.
- Tang, Q., Luo, Y. W., & Wu, X. D. (2023). Research on the evaluation method of agricultural intelligent robot design solutions. *PLoS ONE*, 18(3), e0281554. <https://doi.org/10.1371/journal.pone.0281554>
- Verhoeven, D., Bakker, J., & Veugelers, R. (2016). Measuring technological novelty with patent-based indicators. *Research Policy*, 45(3), 707–723. <https://doi.org/10.1016/j.respol.2015.11.010>
- Wang, J., & Chen, Y. J. (2019). A novelty detection patent mining approach for analyzing technological opportunities. *Advanced Engineering Informatics*, 42(100), 941. <https://doi.org/10.1016/j.aei.2019.100941>
- Wang, Z., Zhang, J., & Feng, J., et al. (2014). Knowledge graph embedding by translating on hyperplanes. In: *Proceedings of the AAAI conference on artificial intelligence*. <https://doi.org/10.1609/aaai.v28i1.8870>
- Wang, Z., Wang, K., Liu, J., et al. (2022). Measuring the innovation of method knowledge elements in scientific literature. *Scientometrics*, 127(5), 2803–2827. <https://doi.org/10.1007/s11192-022-04350-5>
- Wang, Z., Zhang, H., Chen, J., et al. (2023). Measuring the novelty of scientific literature through contribution sentence analysis using deep learning and cloud model. *SSRN*. <https://doi.org/10.2139/ssrn.4360535>
- Yoon, B., Park, I., Yun, D., et al. (2019). Exploring promising vacant technology areas in a technology-oriented company based on bibliometric analysis and visualisation. *Technology Analysis & Strategic Management*, 31(4), 388–405. <https://doi.org/10.1080/09537325.2018.1516864>
- Yoon, J., Seo, W., Coh, B. Y., et al. (2017). Identifying product opportunities using collaborative filtering-based patent analysis. *Computers & Industrial Engineering*, 107, 376–387. <https://doi.org/10.1016/j.cie.2016.04.009>
- Zanella, G., Liu, C. Z., & Choo, K. K. R. (2021). Understanding the trends in blockchain domain through an unsupervised systematic patent analysis. *IEEE Transactions on Engineering Management*. <https://doi.org/10.1109/TEM.2021.3074310>

- Zhang, C. Z., Mayr, P., Lu, W., et al. (2023). Guest editorial: Extraction and evaluation of knowledge entities in the age of artificial intelligence. *Aslib Journal of Information Management*, 75(3), 433–437. <https://doi.org/10.1108/AJIM-05-2023-507>
- Zhang, H., Daim, T., & Zhang, Y. P. (2021). Integrating patent analysis into technology roadmapping: A latent dirichlet allocation based technology assessment and roadmapping in the field of blockchain. *Technological Forecasting and Social Change*, 167(120), 729. <https://doi.org/10.1016/j.techfore.2021.120729>
- Zhao, J., Yang, Y., Zheng, H., et al. (2020). Global agricultural robotics research and development: Trend forecasts. *Journal of Physics: Conference Series*. <https://doi.org/10.1088/1742-6596/1693/1/012227>

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